

HW/Lab 7: Wireless Sensing: Localization

ECSE 4560/6560: Modern Communication Systems

RPI Fall 2025

Goals and Instructions

The goals for this assignment are:

- To understand and apply angle-of-arrival (AoA) geometry for two-antenna and multi-antenna arrays, including the impact of antenna spacing on ambiguity.
- To estimate AoA from relative phase across antennas, and relate this to the localization techniques discussed in lecture.
- To design and implement beam patterns for a uniform linear array (ULA) with different inter-element spacings.

Follow these instructions:

- **Collaboration:** You can discuss concepts, but write your own solutions and code. **Any plagiarism may result in 0 points.**
- **Tools:** We will primarily use MATLAB. Directions use standard MATLAB functions. *Strongly recommended* toolboxes: Signal Processing, Communications. (Base-MATLAB fallbacks are acceptable.)
- **Submission:** Upload a single PDF with your answers and plots to Gradescope before the deadline. Also, compress (into a ZIP file) and upload your ‘lab7 starter code’ folder to a separate Gradescope portal before the same deadline.
- Total 20 points for ECSE 4560 and 25 points for ECSE 6560 students.

Question 1: Angle of Arrival (AoA) True/False

Consider a receiver equipped with *two antennas* separated by distance s . For a far-field source, the incoming wavefront is approximately planar. The resulting phase difference between the two received signals is related to the angle of arrival (AoA) θ by:

$$\frac{\Delta\phi}{2\pi} = \frac{s \cos \theta}{\lambda},$$

where λ is the wavelength and $\Delta\phi$ is the measured phase difference between the two antennas.

For each statement below, indicate whether it is **True or False**, and briefly justify your answer (a few sentences or a simple sketch is sufficient).

- (a) Measuring AoA from only one receiver pair is insufficient for determining the position of the source; AoA estimates from at least two different receiver locations are required for localization.
- (b) If the antenna spacing s is several wavelengths and the system has no noise or multipath, then the two-antenna receiver can determine the source direction with very high accuracy using the equation above.
- (c) The above AoA relationship is unreliable in environments with strong multipath, since reflections distort the phase difference and prevent accurate angle estimation.
- (d) If there is a carrier frequency offset (CFO) between the transmitter and receiver, the receiver must first estimate and compensate for the CFO before using the phase difference to compute AoA.

Question 2: Angle-of-Arrival (AoA) Estimation and Array Ambiguities

Consider a simple 2-antenna uniform linear array (ULA) with inter-element spacing s and a narrowband far-field signal arriving from angle θ (measured from broadside). Let the complex channel responses at the two antennas be h_1 and h_2 , with phase difference

$$\Delta\Phi = \angle(h_2) - \angle(h_1).$$

- (a) **Phase difference model.** Using the geometry of a ULA, explain why the phase difference satisfies

$$\Delta\Phi = \frac{2\pi s}{\lambda} \cos \theta \pmod{2\pi}.$$

List the assumptions needed for this expression to hold.

- (b) **Ambiguity condition.** Show conceptually why the mapping from $\Delta\Phi$ to θ becomes *ambiguous* when the argument of $\cos \theta$ can produce the same phase modulo 2π . Explain why choosing $s > \lambda/2$ introduces angle ambiguities, while choosing $s = \lambda/2$ ensures that

$$-\pi \leq \Delta\Phi \leq \pi,$$

giving a unique AoA estimate.

- (c) **Numerical check.** Let the system operate at 5 GHz. Compute the phase difference $\Delta\Phi$ when

$$\theta = 30^\circ, \quad s = \frac{\lambda}{2}.$$

Then compute the AoA that would produce the *same* $\Delta\Phi$ if the antenna spacing were instead $s = \lambda$. Comment on why this demonstrates the ambiguity problem.

Question 3 (ECSE 6560 only): ArrayTrack Multipath Separation with AoA and TOF

Consider a WiFi access point (AP) equipped with a uniform linear array (ULA) of N antennas with inter-element spacing $s = \lambda/2$. A mobile device sends OFDM packets over a 20 MHz WiFi channel. The AP wants to localize the device by identifying the direct line-of-sight (LoS) path among many multipath components, using an “ArrayTrack-style” approach that combines angle-of-arrival (AoA) and delay information.

- (a) **Why strongest path is not always LoS.** In many indoor environments the strongest path in the AoA spectrum is not necessarily the LoS path. Give two physical reasons why a reflected path can have higher received power than the LoS path. Explain how simply picking the strongest AoA peak can lead to localization errors.
- (b) **Combining AoA and TOF to separate paths.** Suppose that for each AoA peak the AP can also estimate the delay or time-of-flight (TOF) τ using the slope of the OFDM channel phase versus subcarrier frequency. Conceptually outline a three-step procedure that uses:
- (i) the AoA spectrum,
 - (ii) the multipath power-delay profile, and
 - (iii) relative delays between paths,
- to decide which path is the LoS candidate for localization. Your answer should explain how AoA and delay information are jointly used to separate multipath components.
- (c) **Delay resolution.** Consider a 20 MHz WiFi channel. The subcarrier spacing is Δf , and the OFDM symbol duration (useful part) is $T_u = 1/\Delta f$. The AP tries to estimate the delay of a path by fitting a straight line to the unwrapped phase of the channel $H(f)$ across subcarriers, where for a single path

$$H(f) \propto e^{-j2\pi f\tau}.$$

Assuming an effective bandwidth of $B = 20$ MHz, give a rough order-of-magnitude expression for the finest delay resolution $\Delta\tau_{\min}$ that can be achieved from the OFDM channel.

- (d) **Explain.** If the path length difference between two paths is 1 m, comment qualitatively on whether a 20 MHz WiFi system can reliably distinguish their delays using this method, and thus separate them in time.

Programming Questions

The following questions requires writing Matlab code for plots, analysis and explanation.

Question 4: Wireless Channels and Phase Stability

Consider a transmitter (TX) sending a complex symbol x to a receiver (RX). The wireless channel applies attenuation and phase rotation, modeled as a complex scalar h . The received signal is

$$y = hx.$$

In free space, a purely propagation-based channel may be approximated by

$$h \propto \frac{1}{d} e^{-j2\pi d/\lambda},$$

but in practice the measured channel also includes hardware-induced offsets (e.g., frequency offset, sampling clock mismatch, and phase noise), as discussed in class.

For this problem, we collected real measurements using a single-antenna transmitter and a two-antenna receiver, as shown in Fig. 1. The transmitter sent 100 WiFi packets at a narrowband center frequency of 5.5 GHz. The file `h_static.mat` contains:

- **h**: a 100×2 complex matrix where each row is a packet and the two columns correspond to the two receive antennas,
- **t**: a 100×1 vector containing the timestamp (in ms) of each packet.

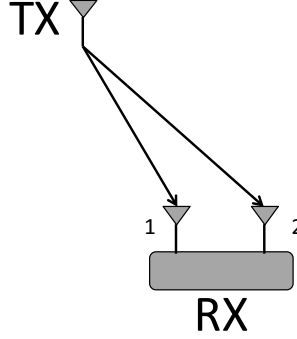


Figure 1: Experimental setup.

Answer the following:

- Plot the channel phase on antenna 1 and antenna 2.** Load the file, compute $\angle(h(:, 1))$ using `angle()`, and plot the phase versus time. Include the figure in your solution.
- Explain the observed phase drift.** Even though the TX and RX are static, the phase on antenna 1 changes over time. Briefly explain which hardware impairments cause this drift.
- Plot the relative phase between antennas.** Compute and plot the phase of the ratio

$$\frac{h(:, 1)}{h(:, 2)}$$

- Interpretation of relative phase stability.** Does the phase of $h(:, 1)/h(:, 2)$ remain approximately constant? Explain why (or why not), referring to the fact that both antennas share the same receiver hardware.
- Estimate angle of arrival (AOA) from relative phase.** Assume the two receive antennas form a uniform linear array (ULA) with spacing $S = \lambda/2$, where λ is the wavelength at 5.5 GHz.

Using your relative phase measurements, compute a single AoA estimate $\hat{\theta}$ (in degrees) by averaging the unwrapped phase of the ratio $h(:, 1)/h(:, 2)$ over all packets to obtain an effective $\Delta\phi$. Report the AoA estimate in degree.

Deliverables: Matlab code (You may use Matlab publish options), relevant plots and explanation for each sub-parts.

Question 5: Channel Phase and Motion Detection

In this experiment, the transmitter is mounted on a Roomba robot that moves horizontally (either from left to right or from right to left) in front of a two-antenna receiver, as illustrated in Fig. 2.

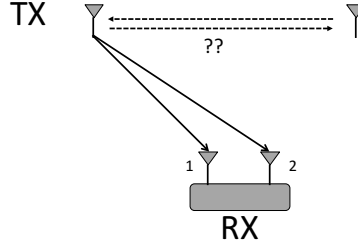


Figure 2: Experimental setup. The transmitter is mounted on a moving Roomba; the receiver has two antennas.

As the Roomba moves, the transmitter periodically sends packets, and the receiver records the complex channel for each packet.

The file `h_move.mat` contains the following variables:

- **h1**: a 200×2 complex matrix of channel measurements for one motion segment (two RX antennas, 200 packets),
- **t1**: a 200×1 vector of timestamps (in ms) corresponding to **h1**,
- **h2**: a 198×2 complex matrix of channel measurements for a second motion segment,
- **t2**: a 198×1 vector of timestamps (in ms) corresponding to **h2**.

In both segments, the Roomba moves at (approximately) constant speed but in potentially different directions (left-to-right vs. right-to-left).

- (a) Load `h_move.mat` in MATLAB and plot the phase of the channel over time for both receive antennas and both motion segments.
Hint: You may find the `angle()` and `unwrap()` functions useful for this step.
- (b) Using only the channel measurements and timestamps, determine the direction of motion of the Roomba in each segment (i.e., which segment corresponds to left-to-right and which to right-to-left). Clearly state your conclusion.
- (c) Briefly describe your approach: which quantity (or quantities) derived from the channel did you examine, and how did that reveal the direction of motion? Do *not* just state the final directions; explain the reasoning with appropriate plots.

Deliverables: Matlab code (You may use Matlab publish options), relevant plots and explanation for each sub-parts.

Question 6: Beam Pattern in a ULA

Consider a uniform linear array (ULA) with $N = 8$ receive antennas, arranged along a line with inter-element spacing s . A narrowband far-field signal of wavelength λ impinges on the array from an angle of arrival (AoA) $\theta_0 = 90^\circ$, which we treat as the “broadside” direction.

For an arbitrary angle $\theta \in [0^\circ, 180^\circ]$, the array steering vector is defined as:

$$\mathbf{a}(\theta) = \begin{bmatrix} 1 \\ e^{-jks \cos \theta} \\ e^{-j2ks \cos \theta} \\ \vdots \\ e^{-j(N-1)ks \cos \theta} \end{bmatrix}, \quad k = \frac{2\pi}{\lambda}.$$

(a) **MATLAB beam pattern for different spacings.**

Let $\lambda = 1$ (normalized) so that only the ratio s/λ matters. You will examine four different inter-element spacings:

$$s \in \left\{ \frac{\lambda}{4}, \frac{\lambda}{2}, \lambda, 2\lambda \right\}.$$

Write a MATLAB script that:

- Sets $N = 8$, $\theta_0 = 90^\circ$, and defines an angle grid θ from 0.1° to 180° with a small step.
- For each spacing s in the set above:
 - Constructs the steering vector $\mathbf{a}(\theta_0)$ and chooses $\mathbf{h} = \mathbf{a}(\theta_0)$. Visualize that $\mathbf{h}$ models the channel phase measurements at N antennas.
 - For each angle θ on the grid, constructs $\mathbf{a}(\theta)$ and computes the beam pattern

$$P(\theta) = \left| \mathbf{h}^H \mathbf{a}(\theta) \right|^2,$$

- Normalizes $P(\theta)$ by its maximum value and converts it to dB (e.g., $10 \log_{10}(\cdot)$), and optionally clips the minimum to -40 dB for plotting.
- Produces a single figure with a 2×2 grid of subplots, one subplot for each spacing s . In each subplot, plot the normalized beam pattern $P(\theta)$ (in dB) versus θ (in degrees), and clearly label which spacing s is being shown.
- A starter code is given and for validation, the expected figure is shown in Figure 3. Replace the ‘beampattern.p’ code with appropriate ‘beampattern.m’ function with your implementation.

(b) **Ambiguity and grating lobes.**

Using your beam pattern plots, identify for which values of s the array response exhibits *ambiguity* in the sense that there are multiple angles θ with strong main-lobe-like peaks (grating lobes) comparable to the desired broadside response at $\theta_0 = 90^\circ$.

Explain in a few sentences why increasing the spacing beyond $s = \lambda/2$ causes such ambiguities.

Deliverables: Matlab code, plots in (a) and answer to part (b).

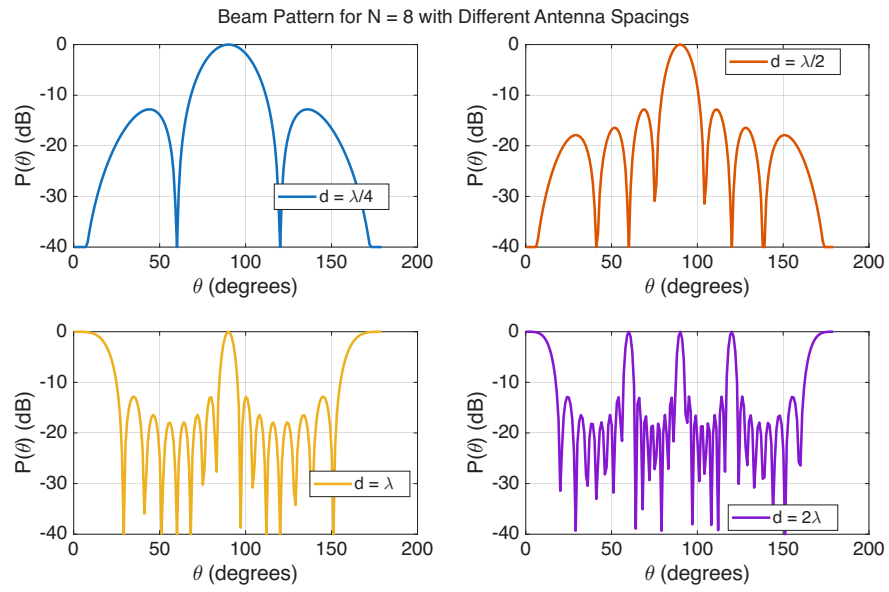


Figure 3: Expected figure for Problem 3.